NOISE WHILE OPERATING A/C SYSTEM [FULL-AUTO AIR CONDITIONER]

id0703c1701000

DESCRIPTION	• Noise from magnetic clutch, A/C compressor, hose or refrigerant line
POSSIBLE CAUSE	• Magnetic clutch operation noise • A/C compressor vane noise • A/C compressor slippage noise • Hose or refrigerant line interference noise

* : If there is gas leakage, air enters the A/C system. The desiccant within the receiver/drier absorbs the moisture from the air and becomes saturated. If the A/C system is used in this condition, the inside of the A/C compressor will begin to rust due to this moisture, which may cause lock up or noise to occur. Therefore, replacement of the receiver/drier is necessary.

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT A/C COMPRESSOR VANE FOR NOISE** • Is there a jingling, popping, beeping, or buzzing sound (A/C compressor vane noise)?	Yes	Go to Step 5.
		No	Go to the next step.
2	**INSPECT A/C COMPRESSOR FOR SLIPPAGE NOISE** • Is there a squeaking or whirling sound (A/C compressor slippage noise)?	Yes	Go to Step 14.
		No	Go to the next step.
3	**INSPECT A/C COMPRESSOR FOR INTERFERENCE NOISE** • Is there a rattling or vibrating sound (interference noise)?	Yes	Go to Step 18.
		No	Go to the next step.
4	**INSPECT MAGNETIC CLUTCH FOR OPERATION NOISE** • Is there a clicking sound (magnetic clutch operation noise)?	Yes	Adjust clearance between pressure plate of magnetic clutch and A/C compressor pulley, then go to Step 19. (See MAGNETIC CLUTCH ADJUSTMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See MAGNETIC CLUTCH ADJUSTMENT [SKYACTIV-D 1.5].)
		No	Condition is normal. (Recheck malfunction symptoms.)
5	**INSPECT A/C COMPRESSOR FOR PERIOD NOISE OCCURS** • Is noise heard continuously for **more than 3 s** after A/C compressor turns on?	Yes	Go to the next step.
		No	Condition is normal. (Noise occurs for **2—3 s** immediately after A/C compressor turns on.)
6	**INSPECT IDLE SPEED** • Inspect idle speed. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ENGINE TUNE-UP [SKYACTIV-D 1.5].) • Is it normal?	Yes	Go to the next step.
		No	Follow the repair instructions indicated in the “ENGINE” section, then go to Step 19.
7	**INSPECT REFRIGERANT AMOUNT** • Inspect refrigerant amount. (See REFRIGERANT CHARGING.) • Is it normal?	Yes	Go to Step 10.
		No	Go to the next step.

Step	Inspection		Action
8	**INSPECT REFRIGERANT LINES** • Inspect refrigerant lines. — Is piping free of damage and cracks? — Are piping connections free of oil grime? (Visual inspection) — Are piping connections free of gas leakage? — Are piping installation points on condenser free of gas leakage? — Are piping installation points on receiver/ drier free of gas leakage? — Are piping installation points on A/C compressor free of gas leakage? — Are piping installation points on A/C unit free of gas leakage? — Perform gas leak inspection using gas leak tester. • Are the above items normal?	Yes	Go to the next step.
		No	If piping or A/C component(s) is damaged or cracked, replace then go to Step 19. If there is gas leakage, repair or replace connection and replace condenser*, then go to Step 19. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See EXPANSION VALVE REMOVAL/INSTALLATION.) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9	**INSPECT EVAPORATOR PIPING CONNECTIONS IN A/C UNIT FOR GAS LEAKAGE** • Are piping connections for evaporator in A/C unit free of gas leakage? (See REFRIGERANT CHARGING.)	Yes	Adjust refrigerant amount to specified level, then go to Step 19. (See REFRIGERANT CHARGING.)
		No	If piping is damaged or cracked, replace then go to Step 19. If there is gas leakage, repair or replace connection and replace condenser*, then go to Step 19. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See EXPANSION VALVE REMOVAL/INSTALLATION.) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10	**VERIFY IF MALFUNCTION IS IN COMPRESSOR OIL OR ELSEWHERE** • Add **20 ml {20 cc, 0.68 fl oz}** of compressor oil. • Is noise heard when racing engine?	Yes	Go to the next step.
		No	Troubleshooting completed. Explain repair to customer.
11	**VERIFY IF MALFUNCTION IS IN A/C COMPRESSOR OR ELSEWHERE** • Drain compressor oil. (See A/C COMPRESSOR REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) • Is it contaminated with metal particles?	Yes	Go to the next step.
		No	Replace A/C compressor, then go to Step 19. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Step	Inspection		Action
12	**VERIFY IF MALFUNCTION IS SOMEWHERE IN A/C SYSTEM OR ELSEWHERE** • Is compressor oil whitish and mixed with water?	Yes	Replace entire A/C system (excluding heater), then go to Step 19. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See RECEIVER/DRIER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See EXPANSION VALVE REMOVAL/INSTALLATION.) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
13	**INSPECT A/C COMPRESSOR OIL** • Is compressor oil darker than normal and contaminated with aluminum chips?	Yes	Replace A/C compressor and condenser, then go to Step 19. (Since A/C compressor may be worn and receiver/drier may be restricted, replacement of receiver/drier is necessary.) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Condition is normal. Recheck malfunction symptoms.
14	**VERIFY IF MALFUNCTION IS IN A/C COMPRESSOR OR ELSEWHERE** • Is noise heard immediately after A/C compressor is stopped?	Yes	Replace A/C compressor, then go to Step 19. (A/C compressor discharge valve left open) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
15	**INSPECT DRIVE BELT** • Inspect the drive belt. (See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DRIVE BELT INSPECTION [SKYACTIV-D 1.5].) • Is it normal?	Yes	Go to the next step.
		No	Replace the drive belt, then go to Step 19. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
16	**INSPECT DRIVE BELT CONDITION** • Is drive belt worn? • Does it have foreign matter imbedded in it, or have oil on it?	Yes	Remove obstruction, remove oil, or replace drive belt, then go to Step 19.
		No	Go to the next step.
17	**INSPECT MAGNETIC CLUTCH** • Inspect the magnetic clutch. (See MAGNETIC CLUTCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See MAGNETIC CLUTCH INSPECTION [SKYACTIV-D 1.5].) • Is it normal?	Yes	Replace A/C compressor (excluding pressure plate, A/C compressor pulley, and stator), then go to Step 19. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Replace magnetic clutch, then go to Step 19. (See MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY [SKYACTIV-D 1.5].)

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18	**VERIFY IF MALFUNCTION IS IN A/C COMPRESSOR OR REFRIGERANT LINES** • Is noise emitted from A/C compressor?	Yes	Visually inspect A/C compressor, replace appropriate parts if necessary, then go to the next step.
		No	If noise is due to refrigerant lines, repair detached or missing clips, tighten loose bolts, then go to the next step.
19	**VERIFY THAT MALFUNCTION SYMPTOM DOES NOT RECUR AFTER REPAIR** • Has A/C compressor noise stopped?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.